

Reproduce HOL blocking with default scheduler

argowf yaml: @spark-pi-workflow.yaml

1. simulate a large long-running spark job which consumes all(almost) resources:

```
argo submit spark-pi-workflow.yaml \  
  -p image=apache/spark:3.5.7 \  
  -p partitions=1000000 \  
  -p namespace=yunikorn-spark-submit \  
  -p executors=16
```

screenshot:

```

● → spark-submit git:(main) x argo submit spark-pi-workflow.yaml \
  -p image=apache/spark:3.5.7 \
  -p partitions=1000000 \
  -p namespace=yunikorn-spark-submit \
  -p executors=16
Name: spark-pi-wf-vg2lb
Namespace: yunikorn-spark-submit
ServiceAccount: unset (will run with the default ServiceAccount)
Status: Pending
Created: Sun Jan 04 17:02:02 +0800 (now)
Progress:
Parameters:
  image: apache/spark:3.5.7
  partitions: 1000000
  namespace: yunikorn-spark-submit
  executors: 16
  queue: root.default
  scheduler: default-scheduler
● → spark-submit git:(main) x kgp

```

NAME	READY	STATUS	RESTARTS	AGE
pythonpi-c0b3049b883e1f04-exec-1	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-10	1/1	Running	0	84s
pythonpi-c0b3049b883e1f04-exec-11	1/1	Running	0	84s
pythonpi-c0b3049b883e1f04-exec-12	1/1	Running	0	83s
pythonpi-c0b3049b883e1f04-exec-13	0/1	Pending	0	83s
pythonpi-c0b3049b883e1f04-exec-14	0/1	Pending	0	83s
pythonpi-c0b3049b883e1f04-exec-15	0/1	Pending	0	83s
pythonpi-c0b3049b883e1f04-exec-16	0/1	Pending	0	82s
pythonpi-c0b3049b883e1f04-exec-2	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-3	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-4	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-5	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-6	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-7	1/1	Running	0	85s
pythonpi-c0b3049b883e1f04-exec-8	1/1	Running	0	84s
pythonpi-c0b3049b883e1f04-exec-9	1/1	Running	0	84s
spark-pi-930fdc9b883e03f9-driver	1/1	Running	0	96s
spark-pi-wf-vg2lb	2/2	Running	0	98s

```

● → spark-submit git:(main) x argo submit spark-pi-workflow.yaml \

```

2. simulate a small short spark job, which we expect low latency:

```

-p image=apache/spark:3.5.7 \
-p partitions=10000 \
-p namespace=yunikorn-spark-submit \
-p executors=1

```

screenshot:

```

➔ spark-submit git:(main) argo submit spark-pi-workflow.yaml \
  -p image=apache/spark:3.5.7 \
  -p partitions=10000 \
  -p namespace=yunikorn-spark-submit \
  -p executors=1
Name: spark-pi-wf-qkfx5
Namespace: yunikorn-spark-submit
ServiceAccount: unset (will run with the default ServiceAccount)
Status: Pending
Created: Sun Jan 04 17:03:54 +0800 (now)
Progress:
Parameters:
  image: apache/spark:3.5.7
  partitions: 10000
  namespace: yunikorn-spark-submit
  executors: 1
  queue: root.default
  scheduler: default-scheduler
➔ spark-submit git:(main) kubectl get pods --sort-by=.metadata.creationTimestamp
NAME                                READY   STATUS    RESTARTS   AGE
spark-pi-wf-vg2lb                   2/2     Running   0           4m47s
spark-pi-930fdc9b883e03f9-driver    1/1     Running   0           4m45s
pythonpi-c0b3049b883e1f04-exec-3    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-1    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-7    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-6    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-5    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-4    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-2    1/1     Running   0           4m34s
pythonpi-c0b3049b883e1f04-exec-9    1/1     Running   0           4m33s
pythonpi-c0b3049b883e1f04-exec-8    1/1     Running   0           4m33s
pythonpi-c0b3049b883e1f04-exec-11   1/1     Running   0           4m33s
pythonpi-c0b3049b883e1f04-exec-10   1/1     Running   0           4m33s
pythonpi-c0b3049b883e1f04-exec-15   0/1     Pending   0           4m32s
pythonpi-c0b3049b883e1f04-exec-14   0/1     Pending   0           4m32s
pythonpi-c0b3049b883e1f04-exec-13   0/1     Pending   0           4m32s
pythonpi-c0b3049b883e1f04-exec-12   1/1     Running   0           4m32s
pythonpi-c0b3049b883e1f04-exec-16   0/1     Pending   0           4m31s
spark-pi-wf-qkfx5                   2/2     Running   0           2m55s
spark-pi-3b4d599b884046c3-driver    0/1     Pending   0           108s

```

conclusion:

with default scheduler and limited resources, there will be HOL blocking, large long-running spark jobs consume all resources will block future jobs in the same ns.

Resolve HOL blocking with yunikorn

1. clear all workloads:

```
→ spark-submit git:(main) argo delete --all
Workflow 'spark-pi-wf-qkfx5' deleted
Workflow 'spark-pi-wf-vg2lb' deleted
→ spark-submit git:(main) x kdel $(kubectl get pod -o name | grep spark-pi)
pod "spark-pi-3b4d599b884046c3-driver" deleted from yunikorn-spark-submit namespace
pod "spark-pi-930fdc9b883e03f9-driver" deleted from yunikorn-spark-submit namespace
spark-submit git:(main) x kgp
no resources found in yunikorn-spark-submit namespace.
→ spark-submit git:(main) x █
```

2. install yunikorn and config queues

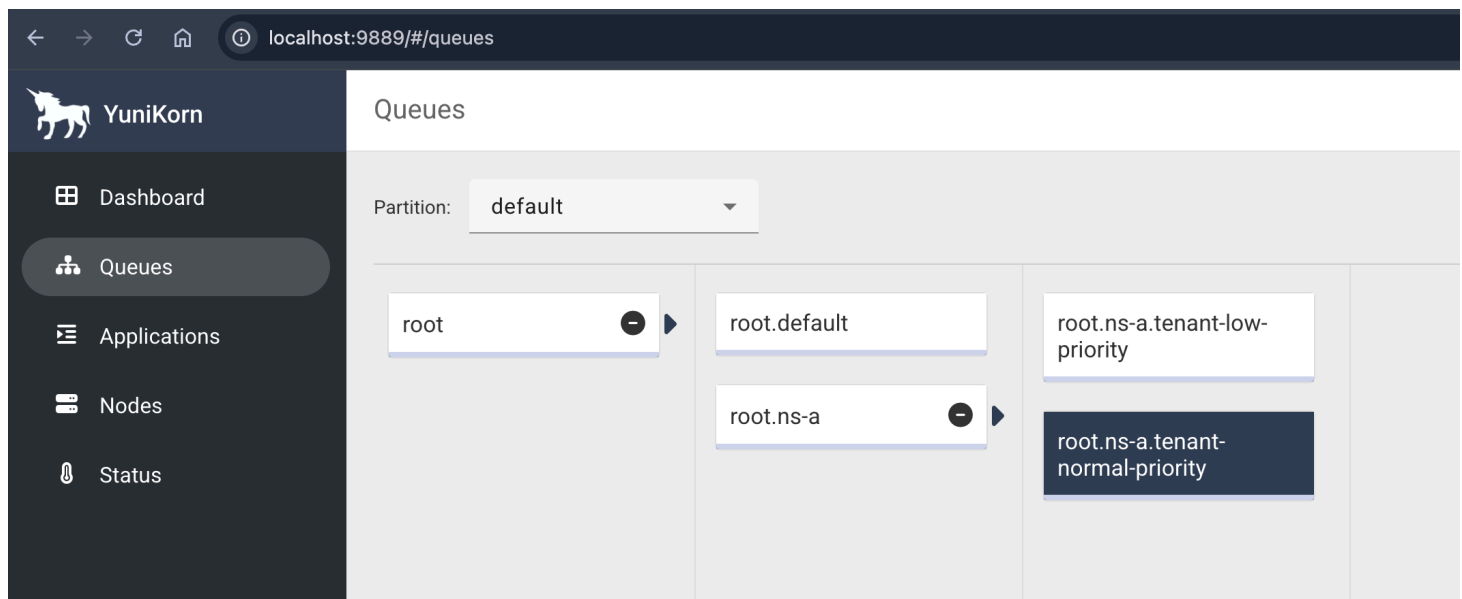
set up 2 queues:

root.ns-a.tenant-low-priority

-- resource: max{memory: 6000Mi, vcore: 1200m}, guaranteed{memory: 3000Mi, vcore: 600m}

root.ns-a.tenant-normal-priority

-- resource: max{memory: 6000Mi, vcore: 1200m}, guaranteed{memory: 3000Mi, vcore: 600m}



3. simulate a large long-running spark job which consume all(almost) allocated resources:

specify yunikorn as the scheduler and the queue with low priority

```

argo submit spark-pi-workflow.yaml \
  -p image=apache/spark:3.5.7 \
  -p partitions=20000 \
  -p namespace=yunikorn-spark-submit \
  -p executors=6 \
  -p queue=root.ns-a.tenant-low-priority \
  -p scheduler=yunikorn

```

```

● → spark-submit git:(main) x argo submit spark-pi-workflow.yaml \
  -p image=apache/spark:3.5.7 \
  -p partitions=20000 \
  -p namespace=yunikorn-spark-submit \
  -p executors=6 \
  -p queue=root.ns-a.tenant-low-priority \
  -p scheduler=yunikorn
Name:          spark-pi-wf-dcx62
Namespace:     yunikorn-spark-submit
ServiceAccount: unset (will run with the default ServiceAccount)
Status:        Pending
Created:        Sun Jan 04 17:59:31 +0800 (now)
Progress:
Parameters:
  image:        apache/spark:3.5.7
  partitions:   20000
  namespace:    yunikorn-spark-submit
  executors:    6
  queue:        root.ns-a.tenant-low-priority
  scheduler:    yunikorn
● → spark-submit git:(main) x k9s -l queue=root.ns-a.tenant-low-priority
NAME                                READY   STATUS    RESTARTS   AGE
pythonpi-ae38439b8872ca81-exec-1   1/1     Running   0           15s
pythonpi-ae38439b8872ca81-exec-2   1/1     Running   0           15s
pythonpi-ae38439b8872ca81-exec-3   1/1     Running   0           15s
pythonpi-ae38439b8872ca81-exec-4   1/1     Running   0           15s
pythonpi-ae38439b8872ca81-exec-5   0/1     Pending   0           15s
pythonpi-ae38439b8872ca81-exec-6   1/1     Running   0           15s
spark-pi-6cf6459b8872a317-driver   1/1     Running   0           29s

```

4. simulate a small short spark job, which we expect low latency:

specify yunikorn as the scheduler and the queue with normal priority, which is higher than the previous low priority queue, so can preempt pods from that queue

```
argo submit spark-pi-workflow.yaml \
  -p image=apache/spark:3.5.7 \
  -p partitions=10000 \
  -p namespace=yunikorn-spark-submit \
  -p executors=2 \
  -p queue=root.ns-a.tenant-normal-priority \
  -p scheduler=yunikorn
```

wait for some momemt will see:

1. some of the exec pods of the low priority job was preempted, and new ones created but in pending state.
2. the normal priority spark job was able to run, because it runs on a different queue with higher priority, and can preempt pods on the low priority queue when it consumes resources less than its guaranteed.

```
● → spark-submit git:(main) x kgp -l queue=root.ns-a.tenant-low-priority
NAME                                READY    STATUS    RESTARTS    AGE
pythonpi-ae38439b8872ca81-exec-10  0/1     Pending   0           17s
pythonpi-ae38439b8872ca81-exec-4    1/1     Running   0           110s
pythonpi-ae38439b8872ca81-exec-5    1/1     Running   0           110s
pythonpi-ae38439b8872ca81-exec-7    0/1     Pending   0           49s
pythonpi-ae38439b8872ca81-exec-8    0/1     Pending   0           49s
pythonpi-ae38439b8872ca81-exec-9    0/1     Pending   0           32s
spark-pi-6cf6459b8872a317-driver    1/1     Running   0           2m4s
● → spark-submit git:(main) x kgp -l queue=root.ns-a.tenant-normal-priority
NAME                                READY    STATUS    RESTARTS    AGE
pythonpi-5e9a539b8873ddb4-exec-1    1/1     Running   0           43s
pythonpi-5e9a539b8873ddb4-exec-2    1/1     Running   0           43s
spark-pi-88724a9b88739e06-driver    1/1     Running   0           60s
○ → spark-submit git:(main) x
```

conclusion:

with yunikorn scheduler and prioritized queues, HOL blocking can be resolved, large long-running spark jobs running on low priority queues can be preempted by jobs running on higher priority queues. thus jobs with higher priority can get resource to execute.